EEI5467_Software Testing and Quality Assurance

Selenium IDE test case documentation
For SoloLearn Website

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Introduction

Overview:

This project focuses on automated testing of a web application Sololearn using Selenium IDE.The goal is to ensure the reliability and functionality of various key features of the web application by creating and executing automated test cases.

Tools and technologies used:

- Extension - Selenium IDE
- Web browser - chrome
- Test script - Java JUnit

Scope:

- Includes the following functionalities:
- Login: Testing valid and invalid login scenarios.
- Quiz Interaction: Answering quiz questions correctly and incorrectly.
- Navigation: Moving between different pages and sections of the website.
- Search: Using the search feature to find courses and handling invalid search queries.

Test cases documentation

Test Case 01
Description: Invalid Login
Preconditions: User is on the login page. User not signed in already
Test Steps:
Open the website.
Navigate to the login page.
Enter invalid username: invalidUSR@gmail.com
Enter invalid password: 123invalid
Click the login button.
Expected Outcome: Error message is displayed indicating invalid login .
Actual Outcome: Error message "Invalid login detected"" is displayed.
Status: Pass. presence of user profile

Test Case 02
Description: Valid Login
Preconditions: User is on the login page.
Test Steps:
Open the website.
Navigate to the login page.
Enter valid username: user@example.com
Enter valid password:********
Click the login button.
Expected Outcome: User is logged in and redirected to the dashboard.
Actual Outcome: User is logged in successfully and redirected to the dashboard.
Status:

Test Case 03
Description: Navigate in navigation bar
Preconditions: User is on any page of the website.
Test Steps:
Click on the links in the navigation menu.
Expected Outcome: User is navigated to the expected pages
Actual Outcome: User is successfully navigated to expected pages
Status: Pass

Test Case 04
Description: Navigate to Courses Page
Preconditions: User is on any page of the website.
Test Steps:
Click on the "Courses" button/link in the navigation menu.
Expected Outcome: User is navigated to the courses page. Displays courses. Navigate to different courses
Actual Outcome: User is successfully navigated to the courses page.Displays courses available
Status: Pass

Test Case 05
Description: Quiz with Correct and incorrect Answer
Preconditions: User is logged in
Test Steps:
Navigate to the courses, quiz section.
Select the correct answer for a question.

Click the check button.
Expected Outcome: Correct answer is accepted, score is updated accordingly.
Actual Outcome: The correct answer is accepted, and the user's score is updated.
Select an incorrect answer for a question.
Click the check button.
Expected Outcome: Incorrect answer is acknowledged

Test Case 06
Description: Search for Q and A
Preconditions: User has logged in.
Test Steps:
Enter a valid search term into the search bar on discussion section .
Click the search button.
Expected Outcome: Relevant search results are displayed.
Actual Outcome: Search results related to Python basics are displayed
Status: Pass

Positive and negative test cases

- Test Case Name: Validate Absence of Error Message on Successful Login
  Description: Verify no error message is displayed upon successful login.
  Preconditions: User is on the login page and has entered valid credentials.
  Test Steps:
  Open the website.
  Navigate to the login page.
  Enter valid username: user@example.com
  Enter valid password:********
  Click the login button.
  Verify that no error message is displayed.
  Expected Outcome: No error message is displayed upon successful login.
  Actual Outcome: No error message

- Description: Validate Presence of Error Message on Failed Login
  Preconditions: User is on the login page.
  Test Steps:
  Open the website.
  Navigate to the login page.
  Enter invalid username: invalidUSR@gmail.com
  Enter invalid password:123invalid
  Click the login button.

Verify that the error message "Invalid username or password" is displayed.
Expected Outcome: Error message "Invalid username or password" is displayed.
Actual Outcome: Error message

- Test Case Name: Validate Course List Presence on Courses Page
  Description: Ensure the course list element is present and visible on the courses page.
  Preconditions: User is on the courses page.
  Test Steps:
    Open the website.
    Navigate to the courses page.

Verify the presence of the course list element.
Expected Outcome: The course list element is present and visible.

Testing Report Summary

The focus of this automation test was on user interactions such as login processes, navigation, quizzes, and search functionalities. Both positive and negative test cases were created to ensure comprehensive coverage.

The testing process involved the following steps:

1. Selenium IDE was installed as a browser extension.
2. A suitable website was chosen based on prior experience with the CS section B.
3. Automated test cases were created for different scenarios including valid and invalid logins, navigating through pages, interacting with elements, and performing searches.
4. The test cases were organized into logical test suites and executed using Selenium IDE.
5. The results of each test case were recorded, noting any deviations from expected outcomes.

Challenges Encountered

1. No adequate prior basic knowledge about selenium
2. Ensuring proper synchronization between the test script and the web application

Lessons Learned

1. Get to know about selenium
2. Organizing test cases into modular suites improves test management, scalability, and maintainability.